OBJECTIVE :

The objective of this task is to develop and execute the given program successfully, understand the underlying concepts, and demonstrate the implementation through code, screenshots, and sample outputs.

SOURCE CODE :

```
#include <stdio.h>

// Recursive function
int fibonacci(int n)
{
    if (n <= 1)
        return n;
    return fibonacci(n - 1) + fibonacci(n - 2);
}

int main()
{
    int n, i;

    printf("Enter the number of terms: ");
    scanf("%d", &n);

    // Iterative Approach
    int a = 0, b = 1, next;
    printf("\nFibonacci Series (Iterative): ");
    for (i = 0; i < n; i++)
    {
        printf("%d ", a);
        next = a + b;
        a = b;
        b = next;
    }

    // Recursive Approach
    printf("\nFibonacci Series (Recursive): ");
    for (i = 0; i < n; i++)
    {
        printf("%d ", fibonacci(i));
    }

    return 0;
}
```

COMPILE RUN/COMMANDS :

The program was compiled and executed using Visual Studio Code with the GCC compiler.

SAMPLE INPUT :

Enter number of terms: 10

SAMPLE OUTPUT-

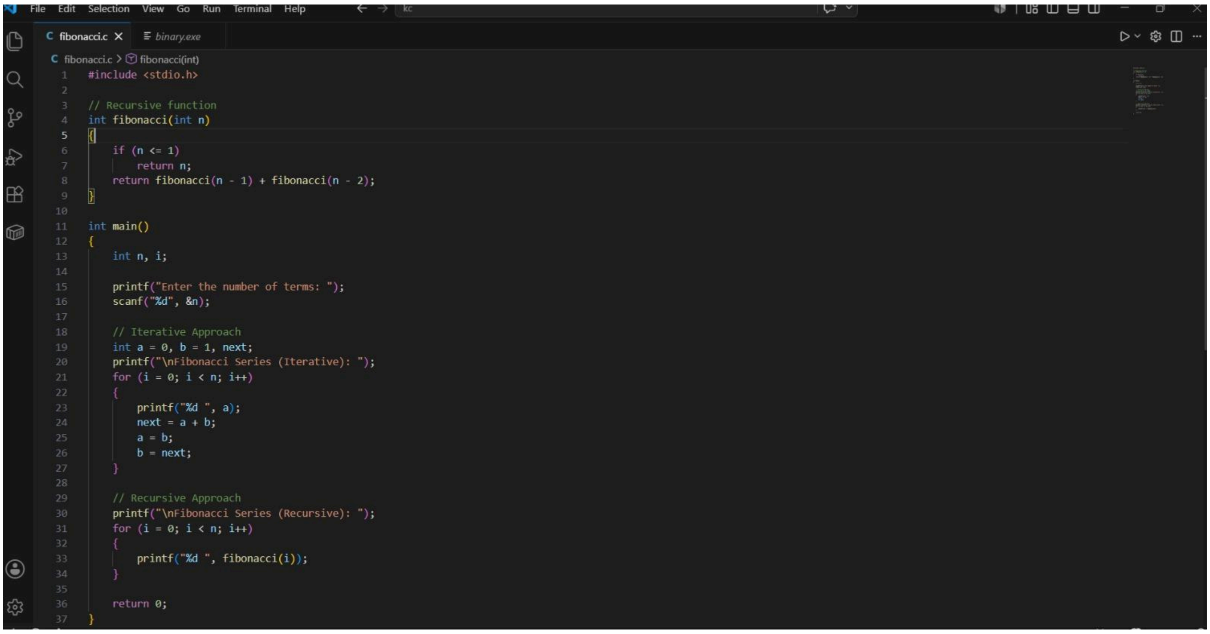
Fibonacci Series(Iterative):
0 1 1 2 3 5 8 13 21 34

Fibonacci Series(Recursive):
0 1 1 2 3 5 8 13 21 34

CONCLUSION :

The Fibonacci Series program was successfully implemented and executed. The output verified the correct generation of Fibonacci numbers. This task helped in understanding loops, variables, and the logic behind the Fibonacci sequence in C programming.

SCREEN SHOTS :



```c
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        next = a + b;
        a = b;
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    }

    // Recursive Approach
    printf("\nFibonacci Series (Recursive): ");
    for (i = 0; i < n; i++)
    {
        printf("%d ", fibonacci(i));
    }

    return 0;
}
```

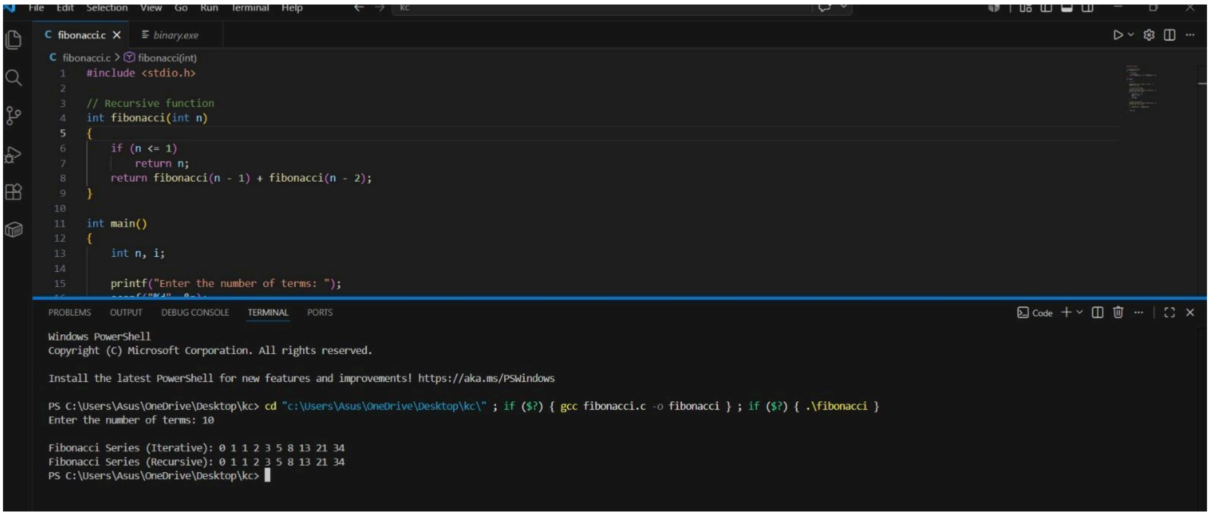

```
Windows PowerShell
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PS C:\Users\Asus\OneDrive\Desktop\kc> cd "c:\Users\Asus\OneDrive\Desktop\kc\" ; if ($?) { gcc fibonacci.c -o fibonacci } ; if ($?) { .\fibonacci }
Enter the number of terms: 10

Fibonacci Series (Iterative): 0 1 1 2 3 5 8 13 21 34
Fibonacci Series (Recursive): 0 1 1 2 3 5 8 13 21 34
PS C:\Users\Asus\OneDrive\Desktop\kc>
```